

EE 175AB: Senior Design

Fall-Winter 2025-2026

Faculty in Charge

Dr. Roman Chomko <chomko@ucr.edu>
Bourns Hall A145

Office Hours: ONLINE by Zoom

<https://ucr.zoom.us/j/91073754852?pwd=RFdLQkhJM2lSVTd0cDEyTjNkcy94UT09>

Instructor Teaching Assistants

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Electrical Engineer

Manglai Zhou, Dept. of Electrical & Computer Engineering, manglai@ece.ucr.edu

Lectures (EE 175B): Mon 5:00 pm – 5:50 pm, Bourns Hall A125

Lab Sections (section numbers can change, but section instructors and prelim meeting times will not):

Section 021: M 6:00 pm – 8:50 pm (Mehrnoosh Ayazi)
Section 022: M 9:00 am – 11:50 am (Shazid Md Islam)
Section 023: F 10:00 am – 12:50 pm (Rajarshi Bhattacharyya)
Section 024: F 2:00 pm – 4:50 pm (Jingzong Zhou)

Specific Course Information

A. Course Objectives

1. Ability to understand the engineering design process.
2. Ability to formulate design specifications and evaluation criteria; determining methodologies and performing solution analyses.
3. Develop skills in project management including organization, teamwork, planning, scheduling, budgeting, and gaining teamwork experience.
4. Develop skills in background research and library techniques such as literature and information searching.
5. Develop technical writing and oral communication skills through proposal and report writing, as well as mid-course and final presentations.
6. Ability to design and conduct experiments, as well as analyze data.
7. Understanding of professional and ethical responsibilities.

8. Obtain a general understanding of engineering economics, marketing, career strategies, and resume preparation.
9. Understand the impact of engineering solutions in a global and societal context.
10. Knowledge of contemporary engineering issues.

B. Student Outcomes Addressed

1. Ability to identify, formulate, and solve complex engineering problems by applying principles of a) engineering, b) science, and c) mathematics.
2. Ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. Ability to communicate effectively with a range of audiences.
4. Ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. Ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. Ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

List of Topics

The Senior Design Project is the culmination of coursework in the bachelor's degree program in electrical engineering or computer engineering. In this comprehensive two-quarter course, students are expected to apply the concepts and theories of electrical engineering or computer engineering to an engineering design project. Detailed written reports, working demonstration, and oral presentations are required.

Eight quarter units of engineering design credit will be granted for the completed project and other required components listed here. It is expected that approximately twelve hours of laboratory (or field) work will be required weekly for satisfactory completion of the project. The design value of these units has been accounted for in the total number of required science and design units necessary for graduation.

Textbook and Related Materials

No textbook is required. All course materials will be posted in iLearn.

ENGR 181W

All students in EE175AB **must take the ENGR 181W** technical writing course (DO NOT take the ENGR 180W alternative). For EE majors, ENGR 181W is the required course and ENGR180W will not be accepted for graduation. ENGR 181W is specifically designed to help you learn how to write professional technical specifications, design documents and test reports. It will start with your final report from EE175AB and help you produce a polished and enhanced final report that you can be proud of when presenting to future employers as evidence of your good work and technical communication capabilities. This can be of great value in your job search and career development. The Board of Industrial Advisors and faculty of the ECE department felt the final reports in past years were not meeting the standard, and ENGR 180W was not sufficient in meeting the needs of EE students and the EE program objectives. ENGR 181W was specifically designed to improve it.

Weekly Class Lectures and Meetings

The format of EE 175A,B one hour lectures will be in-person at scheduled times, or online, if needed (they will be recorded or pre-recorded). You will be informed when lectures will be pre-recorded, and in this case there will be no live lecture. These lectures are intended to provide instruction in topics common to all design projects (design process, technical documentation, engineering economics, ethics, etc.). **Attendance of in-person lectures is a requirement. If, and only if, a lecture is pre-recorded or online, you will be given a short multiple-choice quiz assignment posted in eLearn/Gradescope related to the material of the lecture. If a combined failure on quizzes or missed lecture attendance in any of the two quarters exceeds 3 (three), your final grade will be deducted by half a letter grade regardless of the quality of your project and your project contribution.** To pass the quiz, if given, you will need to answer correctly at least 50% of questions.

In addition, it is expected that each project team meet with the TA on a weekly basis to report and discuss the progress of the project. The weekly meetings may include discussion of weekly progress reports (WPR), discussions of difficulties, problem solving, brief presentations by each team aimed at improving technical presentation skills. **Attendance of weekly lab meetings is mandatory unless allowed by your section TA.** Similar to lecture attendance, **if a combined missed lecture attendance in any of the two quarters exceeds 3 (three), your final grade will be deducted by half a letter grade regardless of the quality of your project and your project contribution.**

ASPECTS OF ONLINE LECTURES (in-person/live or pre-recorded)

Course lectures will be either conducted live in-person or will be pre-recorded and posted in eLearn.

Note on Intellectual Property:

Recording of lectures is hereby strictly prohibited without a consent of instructor

UC policy is that "The University owns the copyright to recordings of classroom lectures, but faculty owns the copyright to their own lecture notes and teaching aids." It is the College's position that studio-based pre-recorded lectures are teaching aids and, therefore, the property of the faculty member who created those lectures. No other faculty member may use them without the explicit approval of the creator. Neither the College nor its representatives will distribute those lectures to others without the explicit approval of the creator of those materials.

Project Topics

Projects span the main electrical engineering areas of the ECE department. You can pick a project suggested by the instructor or you can propose your own topic but it must be approved by your section instructor. In addition, joint projects with other departments may be arranged. General requirements of a successful project are: elements of software, hardware, software/hardware co-design, embedded systems design, PC connectivity, and/or anything else required by your section instructor.

Project Participants

Projects will be completed in small teams with shared responsibility. Each team should have 3 students. Single/two student teams and teams with more than 3 students are not generally allowed. Exceptions require approval by the primary course instructor. Team projects need to be sufficiently complex to allow for an appropriate workload for all team members. Each student will be held responsible for a distinct component of the total team effort.

Project Elements

The senior design projects will include proposal and report writing, experiment design, hardware and software design, test plan and test, analysis of broad impact and ethical issues, among other things. Remember that this is a design course and students must define a *design* project, not a research, nor an evaluation or fabrication project. Each design project must include the following components:

1. **A Clear Technical Design Objective:** Each group must decide upon a design project and get the approval by their section professor by the time of the second lecture. If a team wishes to propose its own project, they must obtain the endorsement of the section professor. Each team should assess the following:
 - Is the design project doable within two quarters?
 - Does the group have the expertise to complete the design, prototype, and testing?
 - Does the group have sufficient funding for the project?
 - Does the group have access to the required test equipment?
 - Is this a design problem (not research, nor fabrication)?
 - Is the project significant enough to be worthy of eight credits (>12 hours/week/person)?
2. **Experiment Design and Feasibility Study** (Required section in Final Report, 5% of final grade) Design and carry out experiments to evaluate the feasibility of project ideas, alternatives, trade-offs and realistic engineering constraints. Analyze the experimental results to prove the feasibility of your project idea and select the best solution to be further developed in the design project.
3. **A Detailed Design Specification:** Describes the functions and quantitatively measurable design objectives, design methods, hardware and software architecture and interfaces, user interface, realistic constraints in terms of time, cost, safety, reliability, social impact, ethics, etc. It must also list and consider the industry standards related to your project, including hardware, protocols, software and tools (e.g., 802.11, RS232, USB, PCI, 3G, API, device drivers, VHDL).
4. **Test Plan** (Required section in Final Report, 5% of final grade): A detailed description of your design of experiments to test and measure whether the final product and each of its components meet the design specifications, and, if not, to test and measure the errors and deviations from specifications.
5. **Design Review Presentation** (Week 10 of the fall quarter, 5% of final grade): Each group must make a Power Point presentation of their design specification and progress to faculty and other students. Requirements of design review presentation will be provided.
6. **Detailed Quantitative Design and Prototype** (To be completed by week 10 of the winter quarter. See the Grading section) Each component of the selected solution and the overall system should be designed and implemented. In most cases, it is necessary to construct a system prototype (or component prototype).
7. **Test Report** (5% of final grade): Carry out the Test Plan you developed to identify how well your final design meet the specifications under the defined constraints, and present the results in this report.
8. **Working Demo and Draft Final Report:** A preliminary working demo of the project is required on week 8 of the winter quarter or at the time required by your section instructor. First completed draft of the final report is due in week 9 of the winter quarter or at the time required by your section instructor.

Your professor will review and mark up draft of your final reports and send it back for improvements and/or re-write.

9. **Final Design Poster Presentation and Final Demo** (Week 9 or 10 of the Winter quarter, 5% of final grade): Each group must make a poster presentation of the final design and show the final working project product to their section instructor and other students at the time designated by your section instructor, ***but no later than one day before the Open House Presentation Event***. The demo should show whether and how design specifications are met. Requirements of final presentation will be provided later.
10. **Final Report** (Final version of the final report is due in week 10 of winter quarter.): The final report must include all the required sections and appendices in a template file, final presentation poster file and video or data of a working demo must be uploaded to iLearn under the main section 001 of the course (an assignment will be created for this purpose).
11. **Open House Presentation Event:** Projects from all sections will have to participate in the Open Demo Day, most likely in the open areas of WCH. Members of the department's Industrial Board of Advisors, representatives from local industries, all professors in ECE and other departments in the college, and all ECE students will be invited. The projects selected need to be well designed, well presented and have a robust working demo. On this day you will demonstrate your ability to communicate with a range of audiences (see program objectives above).

Project Team Selection

There will be three different lab sections with different teaching assistants-instructors (TA). See above for section hours. These sections will be formally established by Wednesday (October 11) of Week 2. ***Do not register to those sections by yourself – your student advisers will do it for you once the EE 175 primary faculty-in-charge submits to them the list of students with their corresponding sections.*** Note that all members of the same team must be eventually registered in the same section. If there is a time conflict with other classes, talk about it to your preferred section instructor whether this time conflict can be waived.

All students will have to enter their projects titles, team members and their section in the spreadsheet files on Google Drive (you have to be logged in to R'Mail to be able to edit).

<https://docs.google.com/spreadsheets/d/1qnZFMrtVZKVnt56WMrNXattLoZG6TOSw/edit?usp=sharing&ouid=100397788337689619358&rtopf=true&sd=true>

If you don't have a project, please check the projects with less than 3 team members, get in touch with them via eLearn email facilities or Piazza to see if your expertise would be a good addition to the project. If you have an interesting project topic in mind, approved by a section instructor or not, please open a topic in Piazza to see, if there are other students who would be interested to work on that project. Remember that all projects must be eventually approved by corresponding section instructors, if the project was not proposed by them. If there is no idea for a project, just indicate your name for any of the empty project and the desired section and see maybe there are other students who would want to work with you on the project that you'll decide upon later during the first two weeks of classes. Just ***make sure that the desired section's instructor can accommodate your hypothetical group*** (the maximum number of groups per instructor is indicated above and in the Google Spreadsheet file).

Critical note: if the number of team members is less than three, and if there are still students without project teams, ***the primary and/or section instructors reserve the right to assign the student to the team even against the will of the rest of the team.***

Project Expense Reimbursements

The department will try, but cannot guarantee, to reimburse your project expenses. At any rate, if departments funds are available the limitations will be as follows:

1. **Reimbursement requests don't exceed \$150 in total**, and are supported by **receipts**;
2. Components are **returned to the ECE department** in completely functional and undamaged form;
3. Reimbursements are requested **no later than May 1, 2025**

Note however that funded projects will be completely re-imbursed.

Accessing Lab Equipment in Chung 126, 128

You will be given access to ECE teaching labs (Chung 126, Chung 128) to work on your project after you pass a lab equipment skills test and a lab/COVID safety exams. You must follow lab rules or your lab access may be revoked.

Weekly Progress Reports (WPR)

In general, each instructor makes weekly assignment in eLearn and Gradescope for the submission of your **individual** (not as a team) weekly progress report which must describe your ***individual*** project contributions. If you worked on some parts as a team, indicate it. An example of a WPR format is the following:

0. File name: ee175_25-26_weekXX_YourName.pdf (or .doc, or docx)

1. Project Title
2. Your name, team member names
3. Week number and dates
4. Detailed description of individual work completed over the past week (schematics, analysis, test data, experiment setup, etc, that is, anything that may be directly copy-pasted into your final report). See the technical merits we are after indicated below.
5. Individual work plan for next week

Note: if for some reason there isn't any work done over any week, and WPR's are required, still submit your WPR with a short description of the reason why there wasn't any work completed (tests, family emergencies, etc). This would be very professional and ethical. Ignorance = Unethical, and professional ethics is something we give credit points for in this course.

Two Required Essays (Submitted separately from Final Report)

1. **Contemporary Engineering Issues** (3% of final grade) Write an essay (500 or more words) on the contemporary engineering principles that were applied to the project, so it produced a solution that met specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors. Everyone on the team must contribute to this essay. Potential contemporary engineering issues related to your project are new technologies, new industry standards, new design methods, new materials, new trends in manufacturing, etc.
2. **Recognition of the need for acquiring and application of new knowledge as needed, using appropriate learning strategies** (3% of final grade) Write an essay (200 or more words) on how you acquired and applied new knowledge as needed, using appropriate learning strategies. In particular, how doing this design project helped you (a) recognize the need and (b) developed the ability in learning new areas and/or subjects. Everyone on the team must contribute to this essay.

Grading

In addition to the deliverables listed above, each project will also be graded on the following:

1. **Design Notes or Project Wiki, Weekly Progress Reports and Lecture Quizzes:** Each student team needs to maintain a design notebook and/or project wiki and weekly progress reports for the duration of their project. Each week, you must show evidence of work completed and progress in the design, implementation and/or testing. (This portion accounts for 7.5% of grade)
2. **Professional Ethics and Responsibility** (7.5% of the final grade but unethical behavior will result in punitive grade deductions): You will be evaluated by your team member(s) and by your section instructor. See the attached evaluation forms on how this is graded.
3. **Technical Merit:** the remaining percentage of the grade are based on the technical and design capabilities, problem solving skills you demonstrated in the design, prototype building, testing, presentation and final report writing.

NOTE: EE 175AB do not have exams.

Grading will be determined by all of the section professors conferring on each project and student. Please note that **grading is individual, not per project team**. That is, for a successful project, a student who contributed most to the success may get an A while another student in the same team who barely did any complex technical work may get a D or F.

Weekly Schedule

Date	Week	Lecture Type	Instructor	Lecture Content
9/26	Week 00	In-Person	All	Introduction, course outline, preliminary issues, requirements and expectations
10/03	Week 01		RC	Introduction to product specification process, engineering design process, understanding prior art, laboratory notebooks, library techniques, literature and information search
10/10	Week 02		RC	Design methodologies, functional block diagram and hardware block diagrams, evaluation of feasibility, problem solving and debug approaches
10/17	Week 03		MZ	Lab skills, exam for equipment use and lab safety for gaining lab access
10/24	Week 04		RC	Experiment design, developing a test plan, collecting data, and evaluation. Realistic design constraints and considerations.
10/31	Week 05		RC	Data Analysis and Error Processing
11/07	Week 06		MZ	Printed circuit board design, layout, and fabrication
11/14	Week 07		RC	Debug Case Study - Hardware and Software
11/21	Week 08		El-Sherief	Systems engineering
11/28	Week 09	Thanksgiving		No lecture Each section to hold project design review in or after this week.

12/05	Week 10			Each section to hold project design review in or before this week.
				Winter Quarter
01/05	Week 11		Career Office	Career choices and strategies, job searches
01/12	Week 12		Career Office	Resume Writing Workshop
01/19	Week 13		RC	Implementation and Debugging of MCU Interfaces - SCI, SPI, I2C OR Engineering product testing requirements, test report requirements. Introduction to and use of industry standards. What is and is not an industry standard. Industry standards vs. de facto standards. Review the important sections of at least two actual industry standards, e.g., 802.11 (standard), SPI or I2C (de facto standard)
01/26	Week 14		RC	Holiday (no lecture)
02/02	Week 15		RC	Understanding professional and ethical responsibility. IEEE Code of Ethics. A required ethics exam will be given at the lecture.
02/09	Week 16		RC	Contemporary Engineering for Solutions with Societal, Environmental and Economic Impacts
02/16	Week 17		TH	Holiday (no lecture)
02/23	Week 18		TH	Writing Technical reports I. Go over the Final Report Template section by section, with instruction on what to write in each section
03/02	Week 19		RC	Writing Technical reports II. Use well-written Technical Reports as examples. OR Patents and intellectual properties: how to apply for a patent, protection of IP. OR Marketing engineering products, entrepreneurship, venture capital and startups Technical project demonstration (to section instructor) Completed draft final report due, if required by section instructor
03/09	Week 20		Open House	Final Project (Slide) Presentation Work with section the instructor to improve final report and essays Senior Design Open House (show your project to the world!) Date: Tuesday, 03/10, 8:00 am – 1:30 pm Essays due
	Finals week			Final version of final report due

RC – Roman Chomko, **MZ** – Manglai Zhou, **TH** – Tofigh Heidarzadeh

Design Review Presentation

On week 9 or 10 of the fall quarter each group is required to make a design review presentation in their sections.

Each group has 10-12 minutes for presentation. Every member of a team must do part of the presentation. Below is a suggested outline for the Power Point slides. Discuss it with your section instructor if there are additional or other requirements for the content.

1. Introduction (purpose, scope, application area/use case)
2. Solution Approach
 - What are requirements and challenges
 - Overview of your system; how your system design meets the requirements
3. System Block Diagram
4. Flow Chart (or functional block diagram) of subsystems
5. Implementation Plan
 - Hardware (what hardware parts to use; specs related to the system requirements)
 - Software (libraries/algorithms to use; UI design; data collection/analysis methods type, etc)
 - Show if you have preliminary results
6. Validation/Verification Plan (metrics to assess performance, expected results, brief testing procedures)
7. Project Management (Gantt chart, work distribution, current progress)

Final Project Presentation and Working Project Demo

On week 8 or 9, per your section instructor's requirement, of the winter quarter, each project group is required to make a poster presentation and demo in their section.

Each group has 10-12 minutes for presentation. Every member of a team must do part of the presentation. Below is the template for the Power Point slides. Your final presentation **MUST** contain the info listed below and it **MUST** include a video of your project demo.

Slide 1: Title, group members, date, contact info

Slide 2: Concept and application of the design: What did you design, technical principles, why this is a meaningful project, what are the intended applications, how is it related to subjects in electrical engineering.

Slide 3: Technical design objectives, be specific and should be measurable (numerical), e.g., response time, frequency or transmission range, SNR, accuracy, false alarm rate, power consumption, capacity, etc.

Slide 4: Final high-level design, show block diagram of the system (SBD).

Slide 5: Final low-level design of the system implementation (schematics, program flow-charts, etc).

Slide 6: Technical challenges, what are the difficult electrical engineering related technical problems you encountered and solved, was there any difficulty that you were unable to solve, and how have you changed your design to get around it?

Slide 7: Major components of the design and implementation, and who contributed to what. Include circuit schematics and program flowcharts on separate slides.

Slide 8: Design considerations:

* Realistic constraints (time, skill, power consumption, weight, size, microcontroller speed, memory size, sampling rate, data rate, costs (this is for weighing the costs of different design decisions. Do NOT simply show a list of parts and their costs), etc.

* What industry standards did your project involve? e.g., 802.11, Bluetooth, I2C, USB, RS232, Zigbee, etc. How using the industry standards affected/constrained your design? (e.g., data rate, voltages, connectors, compatibility with other devices, etc.)

Slide 9: Test report: how did you test the system, what are the test results (in quantitative measures), how close did you come to your design specification?

Slide 10: A real demo or a demo video is required. If your design cannot be demoed at the presentation, you should schedule a demo after the presentation and invite the class to visit

Slide 11: Summary

Slide 12: Acknowledgment of people helped you or provided you with the design concept or ideas

Technical Merits of the Project and the Final Report

The evaluation of technical merits of your project and final report will be based on the reproducibility of your projects, that is, on the clarity of the following:

1. **What the project is all about?** (Provide a clear and brief Executive Summary of your system, a well-annotated and well-explained photo of the system, etc.)
2. **Theoretical basis for the project** (provide the theoretical analysis and associated experiment designs which, when implemented, will confirm the theory of operation)
3. **Design Specifications** (provide a clear outline of features to be expected from the system, expected performance requirements. They must follow from the overall system objectives and goals and the theoretical analysis and/or other considerations)
4. **Implementation Strategy** (describe the System Architecture, show the high-level design, *aka* SBD, etc.)
5. **Actual Implementation** (describe in great detail the low-level design in terms of both hardware and software. Make sure to provide and explain the schematics of all circuits used, show the program and data flow-charts, etc.)
6. **Validation/Verification of Functionality** (provide test results against design specifications, clearly outline conditions for both Pass and Fail, etc)
7. **Future Improvements** (suggest changes, updates and/or alternatives in design in the whole or in parts based on your experiences implementing the project)

Notes of first lecture

1. Projects must be design projects with clearly defined design objectives, not research, experimentation or data collection and analysis.
2. Syllabus with course requirements and final report template in eLearn/Canvas. Use it to check if you covered everything.
3. Attendance of weekly lab sessions is required. See details above. EE 175AB do not have textbooks nor exams.
4. You can pick a project suggested by the instructors or you can propose your own topic, but it must be approved. All students must have decided upon a project and section by the end of the second week.
5. Each project team must be 3 students. Exceptions are approved only in rare cases with strong justifications. No single/two member teams will be allowed since teamwork experience is a course requirement.
6. **All members of the same team will be moved to the same section.**
7. Everyone on the team needs to do approximately equal amount of technical work. The design review, final presentation and final report must clearly indicate who did what. It is not allowed to have just one person do all administrative work, e.g., ordering parts, keeping notes. Each student is graded individually based on his/her contribution.
8. Be a good team player. Teamwork is part of your grade. Everyone must be a reliable and helpful team member. Each member will be evaluated by all other team members at the end of the course.
9. Design specifications and test plans must be **numerical and measurable**.
10. Must do prior art study. Find out what has been done, how other people have done it and what are their advantages and shortcomings. How does your design compare with theirs?
11. Focus is on design using electrical engineering knowledge. Don't spend more than 10% of your effort on mechanical designs.
12. You will be given access to ECE teaching labs (Chung 126, Chung 128) to work on your project after you pass a lab equipment skill test, lab/COVID safety exam. You must follow lab rules or your lab access may be revoked.
13. Mr. Manglai Zhou is the ECE department's Research Engineer, he manages the ECE lab and can provide some parts. He is a very experienced engineer and is available to help you.
14. Instructors will guide you and point you in the right direction, but you must do your own work. Systematic problem solving is a major part of this course. You will be evaluated on how you identify problems, approach them and find solutions.
15. When you are stuck on a problem for too long even after guidance from your section instructor, the instructor may provide a solution to you because we want you to work through the process of getting a design to a functional

prototype. But if your section instructor provides a solution to a problem you are stuck on, points will be deducted from your technical merit score, and thus your grade.

16. Parts of final reports are due throughout the fall and winter quarters, as specified by each section instructor.
17. Document your design process, the alternatives you experimented with, the problems you encountered, how they were solved, and who did what. These all should go into your final report. Do not wait until the end of the course to start writing the final report. Use the Final Report Template as a guide and to document your progress right from the beginning.
18. Final presentation and demo will be on week 8 or 9 of the winter quarter per your section instructor requirement. A complete draft of the final report will be due on week 8 or 9 of winter quarter, the final version of the final report will be due on week 10 or during finals week of winter quarter. Senior Design Open House Presentation Event will be during week 10 of winter quarter subject to COVID/Health restrictions. You must demo a working design to get a grade better than a D.

STUDENT OUTCOMES

Upon completion of this course the student will have to attain all seven ABET student outcomes that will be assessed based on the Course Matrix shown below.

EE 175A/B - Electrical Engineering Senior Design Project										TECHNICAL REQUIREMENT									
EE175A: 4 units, EE175B: 4 units; each quarter one hour consultation, 9 hours laboratory Prerequisites: Senior standing in Electrical Engineering. Under the direction of a faculty member, students (individually or in small teams with shared responsibilities) propose and design electrical engineering devices or systems. Requires an oral report giving details of the																			
Item	OUTCOME-RELATED LEARNING OBJECTIVES	OUTCOMES																	
		1A	1B	1C	2	3	4	5	6	7									
1	Ability to understand the engineering design process.				x														
2	Ability to formulate design specifications and evaluation criteria; determining methodologies and performing solution analyses	x	x	x					x										
3	Develop skills in project management including organization, teamwork, planning, scheduling, and budgeting; gain teamwork experience.					x		x											
4	Develop skills in background research and library techniques such as literature and information searching.									x									
5	Develop technical writing and oral communication skills through proposal and report writing, as well as mid-course and final presentations					x		x											
6	Ability to design and conduct experiments and analyze data				x				x										
7	Understanding of professional and ethical responsibility						x												
8	Obtain a general understanding of engineering economics, marketing, career strategies, and resume preparation									x									
9	Understand the impact of engineering solutions in a global and societal context						x												
10	Knowledge of contemporary engineering issues						x												

OBJECTIVE ADDRESSES OUTCOME:

Outcome 1A: an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering

Outcome 1B: an ability to identify, formulate, and solve complex engineering problems by applying principles of science

Outcome 1C: an ability to identify, formulate, and solve complex engineering problems by applying principles of mathematics

Outcome 2: an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors

Outcome 3: an ability to communicate effectively with a range of audiences

Outcome 4: an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts

Outcome 5: an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives

Outcome 6: an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

Outcome 7: an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Professional Ethics and Responsibility Outcomes

Instructor's Name: _____

Section: _____

TEAM MEMBER EVALUATION FORM

Project Team (name of project) _____

Name of Student Being Evaluated _____

Name of Team Member Evaluating the Student _____

For each attribute, circle the number that best corresponds to how much you feel the attribute was met (1 = very poorly, 5 = very well)

Attribute		Very Poorly			Very Well		
1	Responsibility for performing project duties in a timely manner	1	2	3	4	5	
2	Timely response to e-mails, and /or answering phone calls, and/or team meetings regarding the project status and /or project design decisions	1	2	3	4	5	
3	Use of proper language	1	2	3	4	5	
4	Respect of other team members	1	2	3	4	5	
5	Honesty in stating claims or estimates based on the conducted project research information and/or available data	1	2	3	4	5	
6	Honesty in claiming ability and/or willingness to perform parts of a project	1	2	3	4	5	
7	Honesty in promptly bringing up to attention of other team members factors that may affect individual safety, and/or overall safety of the project, and/or safety, health and welfare of the public, and other factors that might endanger the public or the environment	1	2	3	4	5	
8	Ability to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others	1	2	3	4	5	
9	Helpfulness to team members, and willingness to assist colleagues in their professional development	1	2	3	4	5	

Written Comments:

Department of Electrical Engineering, UCR, Fall/Winter 2025/2026

EE 175 Senior Design
Professional Ethics and Responsibility Outcome

Instructor's Name: _____

Section: _____

STUDENT EVALUATION FORM

Project Team (name of project) _____

Student's Name (to be evaluated) _____

For each attribute, circle the number that best corresponds to how much you feel the course helped students develop that attribute (1 = very poorly, 5 = very well)

INSTRUCTOR'S USE ONLY (please do not evaluate the attributes in this table)

Attribute		Very Poorly			Very Well	
1	Responsibility for performing project duties in a timely manner	1	2	3	4	5
2	Prompt response to comments, remarks and/or requests for information	1	2	3	4	5
3	Results of the Ethics Test given after a lecture on Engineering Ethics	1	2	3	4	5
4	Ethics essay in the Final Report	1	2	3	4	5
5	Responsibility and faithfulness in project status reporting	1	2	3	4	5
6	Being a good team player, responsive to teammate requests, being helpful to teammate	1	2	3	4	5